

# IMO 2026 — Problem 6

Solution by Kimi K3

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**Source:** `results/kimi-k3/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round run: earlier-round audit logs in `scratch/`.

## Problem

Let  $a_1, a_2, a_3, \dots$  be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers  $n$ , the number  $a_{n+1}$  is the smallest positive integer greater than  $a_n$  such that  $\gcd(a_{n+1}, a_i) > 1$  for every  $i = 1, 2, \dots, n$ . Prove that there exist positive integers  $T$  and  $L$  such that  $a_{n+T} = a_n + L$  for every positive integer  $n$ . (Note that  $\gcd(x, y)$  denotes the greatest common divisor of positive integers  $x$  and  $y$ .)

## Solution document

### Status

partial — Steps 1–3 and the reduction are complete and rigorous; the combinatorial core theorem (Step 4) has a confirmed gap at its final step (infinite minimal transversals), flagged by independent review. Closing that gap is the sole remaining task. Plan: prove the *compactness* of the transversal hypergraph of  $\mathcal{C}$  (every transversal contains a finite transversal) using the greedy/killing structure (number theory), which is exactly what the broken step needs. Detailed strategy below; first steps of the number-theoretic argument already derived (see “Gap analysis and repair plan”).

### Approaches tried

- **Minimal-hitting-set (transversal) framework** (session 1):  $S_n = \{m : \gcd(m, a_i) > 1 \ \forall i \leq n\}$ ;  $H_n =$  minimal transversals of  $\{\text{supp}(a_1), \dots, \text{supp}(a_n)\}$ ;  $a_{n+1} =$  successor of  $a_n$  in  $S_n = \min_{h \in H_n}$  next multiple of  $h$ . Gap bound  $a_{n+1} - a_n \leq \text{rad}(a_1)$ ; Lemmas A–D (min-value permanent exists; disjoint pairs die; pure-power terms; permanents = radicals of pure-power terms). Empirics: shrinkage of  $H_n$  always stops; stable  $H$  pairwise intersecting;  $a_{n+T} = a_n + L$  verified for 100+ values of  $a_1$ .
- **Self-dual intersecting clutter** (session 2): permanents (= minimal elements of  $S_\infty$ ) form an antichain  $\mathcal{C}$  of finite prime-sets that is pairwise intersecting and self-dual-for-finite-transversals. Abstract theorem “antichain + intersecting + self-dual finite” proved — BUT its proof applies self-duality to a minimal transversal of the infinite set  $I \cup Z_\omega$ , and in the

application self-duality is only verified for FINITE transversals ( $\pi(T)$  is undefined for infinite  $T$ ). Moreover, in the relevant case (Case B at every stage) that minimal transversal is PROVABLY infinite: any finite transversal  $\subseteq I \cup Z_\omega$  lies in some finite  $I \cup Z_s$ , which would have triggered Case A at stage  $s + 1$ . **This is the open gap.**

- **(THIS SESSION) Repair via compactness from the killing property.** Reformulate greedy as enumeration + killing (K); derive that every transversal of  $\mathcal{C}$  contains a finite transversal (compactness), which repairs Step 4 (equivalently proves (iii) for infinite transversals: none exist). Key derived fact (C1) below; contradiction still being assembled.

## Current best

Complete except for the compactness crux. Established rigorously (see Full proof): (1) the sequence is exactly the increasing enumeration of  $S_\infty \cap [a_1, \infty)$ ; (2) if  $S_\infty$  has finitely many minimal elements, the conclusion follows with  $L = \text{lcm of minimals}$ ,  $T = \text{count per period}$ ; (3) the minimal elements are squarefree, and their supports  $\mathcal{C}$  form an antichain of finite prime sets that is pairwise intersecting and self-dual for finite transversals; (4a) peeling process: if  $\mathcal{C}$  is infinite, one constructs  $t \in \mathcal{C}$ , infinite  $\mathcal{C}_I$ , distinct primes  $Z_\omega = \{z_1, z_2, \dots\}$  outside  $t$  with  $\mathcal{F}_s = \{e \in \mathcal{C}_I : e \supseteq Z_s\}$  infinite for all  $s$ , members  $g_s$  disjoint from  $I \cup Z_s$  with  $z_{s+1} \in g_s$ , and  $I \cup Z_\omega$  a transversal of  $\mathcal{C}$  with no finite sub-transversal. **Remaining:** show number theory (the killing property) forbids such an infinite transversal-without-finite-subtransversal.

## Gap analysis and repair plan (this session's working notes — will be folded into the proof once complete)

**Reformulation of the greedy rule.** The greedy definition is EQUIVALENT to:  $(a_n)$  is the increasing enumeration of  $S_\infty \cap [a_1, \infty)$  AND

(K) every integer  $x \notin S_\infty$ ,  $x > a_1$ , is coprime to some term  $a_i < x$ .

[Proof: greedy enumeration (Step 1) and any  $x \in (a_n, a_{n+1})$  fails the gcd condition with some  $i \leq n$ , i.e.  $\gcd(x, a_i) = 1$ ,  $a_i \leq a_n < x$ . Conversely, enumeration + (K): if  $y \in (a_n, a_{n+1})$  had  $\gcd(y, a_i) > 1$  for all  $i \leq n$ , then  $y \notin S_\infty$  (enumeration is complete), so (K) gives a term  $a_j < y$  coprime to  $y$ ;  $a_j < y < a_{n+1}$  forces  $j \leq n$  — contradiction. So the min in the greedy rule equals  $a_{n+1}$ .]

**Setup for the contradiction.** Suppose  $\mathcal{M}$  (equivalently  $\mathcal{C}$ ) infinite. The peeling (valid, purely combinatorial — antichain + intersecting only) yields  $t \in \mathcal{C}$ ,  $\emptyset \neq I \subseteq t$ ,  $\mathcal{C}_I = \{e : e \cap t = I\}$  infinite, distinct primes  $z_1, z_2, \dots \notin t$  with  $\mathcal{F}_s$  infinite,  $g_s \in \mathcal{C}$  disjoint from  $I \cup Z_s$ ,  $z_{s+1} \in g_s$ , and  $W := I \cup Z_\omega$  a transversal of  $\mathcal{C}$  such that no finite subset is a transversal. **KEY NUMBER-THEORETIC FACTS:** - For finite  $F \subseteq W$ :  $F$  contains no member of  $\mathcal{C}$  (else that member meets... wait — if  $F \supseteq e \in \mathcal{C}$  then since some  $g$  disjoint from  $F$  exists [no finite sub-transversal] we'd contradict intersecting). Hence  $\pi(F) \notin S_\infty$ . - Applying (K) to  $x = \pi(F)$  (large  $F$ ): some term  $a_i < \pi(F)$  is coprime to  $\pi(F)$ ; any permanent  $t' \subseteq \text{supp}(a_i)$  then satisfies  $t' \cap F = \emptyset$  and  $\pi(t') \leq a_i < \pi(F)$ . Hence

$$(C1) \quad \forall \text{ finite } F \subseteq W \ (\pi(F) > a_1), \exists t' \in \mathcal{C} : t' \cap F = \emptyset, \pi(t') < \pi(F).$$

- Since  $W$  is a transversal, each such  $t'$  contains some  $w \in W \setminus F$ , and  $w \leq \pi(t') < \pi(F)$ .

**Where the contradiction should come from (exploring):** - (i) Iterating (C1) with  $|F| = 1$  gives, for each  $w_0 \in W$ ,  $w_0 > a_1$ , a strictly decreasing chain  $w_0 > w_1 > \dots > w_k$  in  $W$  ending with

$w_k \leq a_1$ . No contradiction yet, but shows  $W$ 's elements  $> a_1$  “descend” to  $\leq a_1$  along permanents. - (ii) Stronger: iterate with  $F_s = \{w_0, \dots, w_s\}$ : get  $w_{s+1} \in W \setminus F_s$ ,  $w_{s+1} < \prod_{i \leq s} w_i$  AND a permanent  $t_{s+1} \ni w_{s+1}$  disjoint from  $F_s$  with  $\pi(t_{s+1}) < \prod F_s$ . Still not a contradiction (growth condition too weak) — UNLESS combined with a lower bound on permanents' elements or with disjointness structure. - (iii) POTENTIAL ROUTE (most promising): use (C1) to show some FIXED finite subfamily of  $\mathcal{C}$  already blocks all of  $W$ ... More precisely: consider  $\mathcal{C}^{<N} := \{t \in \mathcal{C} : \pi(t) < N\}$  — finite for each  $N$ . (C1) says: every finite  $F \subseteq W$  is disjoint from some member of  $\mathcal{C}^{<\pi(F)}$ . Equivalent: the family  $\{t \cap W : t \in \mathcal{C}\}$  is a family of subsets of  $W$  with the finite intersection property w.r.t. complements... Aim: find finite  $F \subseteq W$  meeting every  $t \in \mathcal{C}$  with... hmm — actually what we need: a finite  $F \subseteq W$  that IS a transversal. Compactness equivalent. Idea: order permanents by  $\pi$ -value; use (C1) + the extra structural fact that  $t' \cap F = \emptyset$  AND  $t'$  meets  $W$  AND  $t'$  meets every member (intersecting) to build either a decreasing  $\pi$ -value sequence of permanents with pairwise-disjoint... decreasing positive integers must terminate, at which point the terminal permanent's intersection with  $W$  plus earlier choices gives a finite transversal. Being worked out in scratch/session3-notes.md. - (iv) Alternative: gap bound. Previous session recorded  $a_{n+1} - a_n \leq \text{rad}(a_1)$  (needs re-proof). If gaps are bounded by  $R$ , then any interval of length  $R$  contains a term, so any  $x \notin S_\infty$  is killed by a term in  $(x - R, x)$ . With  $W$  as above and  $\pi(F)$  growing, killing terms live in  $(\pi(F) - R, \pi(F))$  — but permanents below killing terms are  $< \pi(F)$  anyway. Unclear if stronger.

### Full proof (steps that are complete; the compactness lemma is being repaired)

**Notation.** For an integer  $m \geq 2$ ,  $\text{supp}(m)$  = set of prime divisors of  $m$ . For a finite set  $s$  of primes,  $\pi(s) := \prod_{p \in s} p$  (a squarefree integer). A set  $s$  of primes is a *transversal* of a family  $\mathcal{F}$  of sets if  $s \cap F \neq \emptyset$  for every  $F \in \mathcal{F}$ .

#### Step 1: The sequence enumerates $S_\infty \cap [a_1, \infty)$

Let  $S_\infty := \{m \geq 1 : \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}$ .

*Claim.*  $(a_n)_{n \geq 1}$  is the increasing enumeration of  $S_\infty \cap [a_1, \infty)$ .

*Proof.* Every term lies in  $S_\infty$ : for  $i < j$ , the defining property of  $a_j$  (smallest integer  $> a_{j-1}$  with  $\gcd(a_j, a_{i'}) > 1$  for all  $i' \leq j-1$ ) applied with  $i' = i$  gives  $\gcd(a_j, a_i) > 1$ ; also  $\gcd(a_i, a_i) = a_i > 1$ . So any two terms share a factor  $> 1$ , i.e. every term is in  $S_\infty$ ; also  $a_n \geq a_1$  as the sequence is strictly increasing.

Conversely let  $x \in S_\infty$ ,  $x \geq a_1$ , and suppose  $x$  is not a term. Then  $x > a_1$  (else  $x = a_1$  is a term), and since the terms form an infinite strictly increasing sequence of integers, there is a unique  $n$  with  $a_n < x < a_{n+1}$ . By definition  $a_{n+1}$  is the smallest integer  $> a_n$  in  $S_n := \{m : \gcd(m, a_i) > 1 \forall i \leq n\}$ . But  $x \in S_\infty \subseteq S_n$  and  $x > a_n$ , so  $a_{n+1} \leq x$  — contradiction. ■

#### Step 2: Reduction to finiteness of the minimal elements of $S_\infty$

$S_\infty$  is closed under taking multiples. Let  $\mathcal{M}$  be the set of its minimal elements under divisibility. Every  $m \in S_\infty$  is a multiple of some element of  $\mathcal{M}$ : among the divisors of  $m$  lying in  $S_\infty$  (a nonempty finite set) the smallest is a minimal element. Hence  $S_\infty = \bigcup_{h \in \mathcal{M}} h\mathbb{Z}^+$ .

*Claim.* If  $\mathcal{M}$  is finite, the problem's conclusion holds.

*Proof.* Write  $\mathcal{M} = \{h_1, \dots, h_k\}$  and  $L := \text{lcm}(h_1, \dots, h_k) \geq 1$ . Since each  $h_i \mid L$ , for every positive integer  $m$ :  $m \in S_\infty \iff \text{some } h_i \mid m \iff \text{some } h_i \mid m + L \iff m + L \in S_\infty$ . Let  $T := \#(S_\infty \cap [a_1, a_1 + L))$ ; since  $a_1 \in S_\infty$  (Step 1),  $T \geq 1$ . The map  $x \mapsto x + L$  is an order-preserving bijection  $S_\infty \cap [a_1, \infty) \rightarrow S_\infty \cap [a_1 + L, \infty)$ ; iterating, each block  $[a_1 + jL, a_1 + (j+1)L)$  ( $j \geq 0$ ) contains exactly  $T$  elements of  $S_\infty$ , namely  $a_1 + jL, \dots, a_T + jL$ . By Step 1 the terms in that block are exactly  $a_{jT+1}, \dots, a_{jT+T}$ , so  $a_{jT+r} = a_r + jL$  for  $1 \leq r \leq T$ ,  $j \geq 0$ . Writing  $n = jT + r$ :  $a_{n+T} = a_{(j+1)T+r} = a_r + (j+1)L = a_n + L$  for every  $n \geq 1$ . ■

**Step 3: The minimal elements form an intersecting, finite-self-dual antichain of finite sets**

Every  $h \in \mathcal{M}$  is squarefree: if  $p^2 \mid h$  for a prime  $p$ , then  $h/p \in S_\infty$  (since  $\text{supp}(h/p) = \text{supp}(h)$  still meets every  $\text{supp}(a_i)$ ), contradicting minimality. So each  $h \in \mathcal{M}$  has the form  $h = \pi(t)$  for a finite set of primes  $t = \text{supp}(h)$ . For squarefree  $h = \pi(t)$  we have  $h \in S_\infty$  iff  $t$  is a transversal of  $\{\text{supp}(a_i) : i \geq 1\}$ , and  $h$  minimal under divisibility iff  $t$  is minimal under inclusion (a proper divisor of a squarefree  $h$  lying in  $S_\infty$  is exactly a proper sub-transversal). Hence

$$\mathcal{C} := \{\text{supp}(h) : h \in \mathcal{M}\}$$

is exactly the family of minimal (under inclusion) transversals of  $\{\text{supp}(a_i) : i \geq 1\}$ . It is an antichain of finite sets: if  $t \subseteq t'$  are both minimal transversals, then  $t = t'$ .

*Claim 3.1 (intersecting).* Any two members of  $\mathcal{C}$  intersect.

*Proof.* Let  $t \in \mathcal{C}$ . Every multiple of  $\pi(t)$  lies in  $S_\infty$  (its support contains  $t$ , hence meets every  $\text{supp}(a_i)$ ). Choose  $k \geq 1$  with  $\pi(t)^k \geq a_1$ ; by Step 1,  $\pi(t)^k$  is a term  $a_j$ , and  $\text{supp}(a_j) = t$ . Every  $t' \in \mathcal{C}$  is a transversal of  $\{\text{supp}(a_i)\}$ , so  $t' \cap t = t' \cap \text{supp}(a_j) \neq \emptyset$ . ■

*Claim 3.2 (self-dual for finite transversals).* Every FINITE minimal transversal of the family  $\mathcal{C}$  belongs to  $\mathcal{C}$ .

*Proof.* Let  $T$  be a FINITE minimal (under inclusion) transversal of  $\mathcal{C}$ . For each  $i$ ,  $a_i \in S_\infty$  is a multiple of some  $\pi(t)$  with  $t \in \mathcal{C}$ ; then  $t \subseteq \text{supp}(a_i)$ , and  $T$  meeting  $t$  implies  $T$  meets  $\text{supp}(a_i)$ . Hence  $\pi(T) \in S_\infty$ , so  $\pi(T)$  is a multiple of some  $\pi(t')$  with  $t' \in \mathcal{C}$ , i.e.  $t' \subseteq T$ . By Claim 3.1  $t'$  meets every member of  $\mathcal{C}$ , so  $t'$  is a transversal of  $\mathcal{C}$  contained in  $T$ ; minimality of  $T$  gives  $T = t' \in \mathcal{C}$ . ■

**Step 4: Finiteness of  $\mathcal{C}$  — peeling + compactness**

**PART 1 (complete): peeling.** Suppose  $\mathcal{C}$  is infinite. Fix  $t \in \mathcal{C}$ ; by Claim 3.1 every member meets the finite set  $t$ , so  $\mathcal{C} = \bigsqcup_{\emptyset \neq I \subseteq t} \mathcal{C}_I$  with  $\mathcal{C}_I := \{e : e \cap t = I\}$ ; fix  $I$  with  $\mathcal{C}_I$  infinite. Observation (b): if  $g \in \mathcal{C}$  is disjoint from  $I$ , then  $g$  meets every  $e \in \mathcal{C}_I$  outside  $t$  (since  $g \cap e \neq \emptyset$  and  $g \cap e \cap t \subseteq g \cap I = \emptyset$ ). Enumerate  $\mathcal{C}$  and inductively build distinct  $z_1, z_2, \dots \notin t$  with  $\mathcal{F}_s := \{e \in \mathcal{C}_I : e \supseteq Z_s\}$  infinite: at stage  $s+1$ , if some member is disjoint from  $I \cup Z_s$ , take the least-index such  $g_s$ ; by (b),  $g_s$  meets every member of the infinite  $\mathcal{F}_s$  outside  $t$ ; pigeonhole gives  $z_{s+1} \in g_s \setminus t$  lying in infinitely many members of  $\mathcal{F}_s$ ; set  $\mathcal{F}_{s+1} := \{e \in \mathcal{F}_s : z_{s+1} \in e\}$ . If instead NO member is disjoint from  $I \cup Z_s$ , then  $I \cup Z_s$  is a FINITE transversal of  $\mathcal{C}$  — the process stops with a finite transversal.

So EITHER the process stops at a finite transversal  $I \cup Z_s$  of  $\mathcal{C}$ , OR it runs forever; in the latter case  $W := I \cup Z_\omega$  is a transversal of  $\mathcal{C}$  (proof: if  $g$  were disjoint from  $W$ , then  $g$  is eligible at

every stage; the least-eligible choices  $g_s$  have strictly increasing indices [shown in lemmas file], so  $\text{index}(g_s) \rightarrow \infty$ , but all are  $\leq \text{index}(g)$  — contradiction) with NO finite sub-transversal (any finite  $F \subseteq W$  lies in some  $I \cup Z_s$ , and  $g_s$  is disjoint from it).

**PART 2 (THE OPEN GAP): compactness.** THEOREM TO PROVE: *every transversal of  $\mathcal{C}$  contains a finite transversal* (equivalently: no infinite transversal without finite sub-transversal exists). This must use the killing property (K) — purely set-theoretic axioms do NOT suffice (star-like configurations satisfy antichain + intersecting + finite-self-duality... [verification pending, see scratch]). Given PART 2, Step 4 concludes: in the “runs forever” case, Part 2 applied to  $W$  gives a finite transversal  $F \subseteq W$ , contradicting the last clause of Part 1. In the “stops” case,  $I \cup Z_s$  is a finite transversal; a minimal sub-transversal is finite, hence in  $\mathcal{C}$  (Claim 3.2), and is contained in every member of the infinite  $\mathcal{F}_s$ , forcing  $\mathcal{F}_s = \{T\}$  — contradiction. Either way  $\mathcal{C}$  is finite, and Step 2 finishes.

**(K) and its consequence (C1)** [proved in “Gap analysis” above; to be formalized]: every  $x \notin S_\infty$ ,  $x > a_1$ , is coprime to some term  $< x$ ; applied to  $x = \pi(F)$  for finite  $F \subseteq W$ : some permanent  $t'$  has  $t' \cap F = \emptyset$  and  $\pi(t') < \pi(F)$ .

**ASSEMBLY OF CONTRADICTION — IN PROGRESS** (see scratch/session3-notes.md).